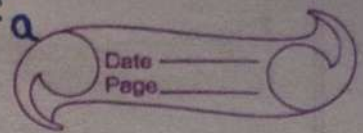


For $\dagger$, $n \cdot a = 0 \Rightarrow n$ is the order of a
For $\times$, $a^n = 1$ & $a \in G$.



Order of an Element of a Group

$$n \cdot a = e$$

The smallest positive integer ' n ' such that $a^n = e$ (where e is the identity element) is considered as the order of an element and it is denoted by $O(a)$ or $|a|$, where $a \in G$.

i.e. ' e '

Note: The order of the identity element is always 1 and $O(a) = O(a^{-1})$.

Q. $G = \{1, -1, i, -i\}$, $\times$. Find the order of all elements?

Solⁿ:

Here, identity element is $e = 1$

$$1^1 = 1 \checkmark$$

$$O(1) = 1$$

$$(-1)^1 = -1 \neq 1 \times$$

$$O(-1) = 2$$

$$(-1)^2 = 1 \checkmark$$

$$i^1 = i \times$$

$$i^2 = -1 \times$$

$$O(i) = 4$$

$$i^3 = -i \times$$

$$i^4 = 1 \checkmark$$

$$(-i)^1 = -i \times$$

$$(-i)^2 = -1 \times$$

$$O(-i) = 4$$

$$(-i)^3 = i \times$$

$$(-i)^4 = 1 \checkmark$$

Q. $G = \{0, 1, 2, 3\}$, $+$. Find the order of 0, 1, 2 & 3?

Solⁿ:

Composition table,

$+$	0	1	2	3
0	0	1	2	3
1	1	2	3	0
2	2	3	0	1
3	3	0	1	2

$[n \cdot a = 0]$

For 0,

$$1 \cdot 0 = 0 \checkmark$$

$$\Rightarrow O(0) = 1$$

For 1,

$$1 \cdot 1 = 1 \times$$

$$2 \cdot 1 = 1 +_4 1 = 2 \times$$

$$3 \cdot 1 = 1 +_4 1 +_4 1 = 2 +_4 1 = 3 \times$$

$$4 \cdot 1 = 1 +_4 1 +_4 1 +_4 1 = 3 +_4 1 = 0 \checkmark$$

$$\Rightarrow O(1) = 4$$

For 2,

$$1 \cdot 2 = 2 \times$$

$$2 \cdot 2 = 2 +_4 2 = 0 \checkmark$$

$$\Rightarrow O(2) = 2$$

For 3,

$$1. 3 = 3 \times$$

$$2. 3 = 3 +_4 3 = 2 \times$$

$$3. 3 = 3 +_4 3 +_4 3 = 2 +_4 3 = 1 \times$$

$$4. 3 = 3 +_4 3 +_4 3 +_4 3 = 1 +_4 3 = 0 \checkmark$$

$$\Rightarrow O(3) = 4$$

Q. $G = \{1, 2, 4, 7, 8, 11, 13, 14\}$, $\times_{15}$. Find the order of 13?

Solⁿ:

Here, $e = 1$.

Composition table,

$\times_{15}$	1	2	4	7	8	11	13	14
1	1	2	4	7	8	11	13	14
2	2	4	8	14	1	7	11	13
4	4	8	1	13	2	14	7	11
7	7	14	13	4	11	2	1	8
8	8	1	2	11	4	13	14	7
11	11	7	14	2	13	1	8	4
13	13	11	7	1	14	8	4	2
14	14	13	11	8	7	4	2	1

$$13^1 = 13$$

$$13^2 = 13 \times_{15} 13 = 4$$

$$13^3 = 13 \times_{15} 13 \times_{15} 13 = 4 \times_{15} 13 = 7$$

$$13^4 = 13^3 \times_{15} 13 = 7 \times_{15} 13 = 1 \checkmark$$

$$\Rightarrow O(13) = 4.$$

Cyclic Group Generators

In a group $(G, *)$, an element a is called generator if $a \in G$ and for every $x \in G$ can be represented using power of a .

For example, $G = \{0, 1, 2, 3\}$ w.r.to $+_4$.

$+_4$	0	1	2	3
0	0	1	2	3
1	1	2	3	0
2	2	3	0	1
3	3	0	1	2

For 0,

$$1 \cdot 0 = 0$$

$$2 \cdot 0 = 0$$

$$3 \cdot 0 = 0$$

For 1,

$$1 \text{ or } 1 \cdot 1 = 1$$

$$2 \cdot 1 = 1 +_4 1 = 2$$

$$3 \cdot 1 = 2 +_4 1 = 3$$

$$4 \cdot 1 = 3 +_4 1 = 0$$

Since, $1 \in G$ and can be used to represent all other elements.

Thus, 1 is a generator or generating element.

Note: A group $(G, *)$ is said to be cyclic group if it contains atleast one generating element.

Q. Prove that the group $G = \{1, 2, 3, 4, 5, 6\}$ w.r.to X_7 is a cyclic group?

Solⁿ:

Composition table,

X_7	1	2	3	4	5	6
1	1	2	3	4	5	6
2	2	4	6	1	3	5
3	3	6	2	5	1	4
4	4	1	5	2	6	3
5	5	3	1	6	4	2
6	6	5	4	3	2	1

For 1,

$$1^1 = 1$$

$$1^2 = 1 \times_7 1 = 1$$

$$1^3 = 1 \times_7 1 = 1$$

$\vdots$

For 2,

$$2^1 = 2$$

$$2^2 = 2 \times_7 2 = 4$$

$$2^3 = 4 \times_7 2 = 1$$

$$2^4 = 1 \times_7 2 = 2$$

$$2^5 = 2 \times_7 2 = 4$$

$$2^6 = 4 \times_7 2 = 1$$

$\vdots$

For 3,

$$3^1 = 3$$

$$3^2 = 3 \times_7 3 = 2$$

$$3^3 = 2 \times_7 3 = 6$$

$$3^4 = 6 \times_7 3 = 4$$

$$3^5 = 4 \times_7 3 = 5$$

$$3^6 = 5 \times_7 3 = 1$$

Here, 3 is a generating element.

Hence, the given group is cyclic.

Q. Find all the generating elements of the group $G = \{1, 2, 4, 7, 8, 11, 13, 14\}$ w.r. to $\times_{15}$?

Solⁿ:

$\times_{15}$	1	2	4	7	8	11	13	14
1	1	2	4	7	8	11	13	14
2	2	4	8	14	1	7	11	13
4	4	8	1	13	2	14	7	11
7	7	14	13	4	11	2	1	8
8	8	1	2	11	4	13	14	7
11	11	7	14	2	13	1	8	4
13	13	11	7	1	14	8	4	2
14	14	13	11	8	7	4	2	1

For 1,

$$1^1 = 1$$

$$1^2 = 1 \times_{15} 1 = 1$$

$$1^3 = 1 \times_{15} 1 = 1$$

$\vdots$

For 2,

$$2^1 = 2$$

$$2^2 = 2 \times_{15} 2 = 4$$

$$2^3 = 4 \times_{15} 2 = 8$$

$$2^4 = 8 \times_{15} 2 = 1$$

$$2^5 = 1 \times_{15} 2 = 2$$

$$2^6 = 2 \times_{15} 2 = 4$$

⋮

For 3,

$$\cancel{3^1 = 3}$$

$$\cancel{3^2 = 3 \times_{15} 3 = 9}$$

$$\cancel{3^3 = 9 \times_{15} 3 = 12}$$

For 4,

$$4^1 = 4$$

$$4^2 = 4 \times_{15} 4 = 1$$

$$4^3 = 1 \times_{15} 4 = 4$$

⋮

For 7,

$$7^1 = 7$$

$$7^2 = 7 \times_{15} 7 = 4$$

$$7^3 = 4 \times_{15} 7 = 13$$

$$7^4 = 13 \times_{15} 7 = 1$$

$$7^5 = 1 \times_{15} 7 = 7$$

⋮

For 8,

$$8^1 = 8$$

$$8^2 = 8 \times_{15} 8 = 4$$

$$8^3 = 4 \times_{15} 8 = 2$$

$$8^4 = 2 \times_{15} 8 = 1$$

$$8^5 = 1 \times_{15} 8 = 8$$

$\vdots$

For 11,

$$11^1 = 11$$

$$11^2 = 11 \times_{15} 11 = 1$$

$$11^3 = 1 \times_{15} 11 = 11$$

$\vdots$

For 13,

$$13^1 = 13$$

$$13^2 = 13 \times_{15} 13 = 4$$

$$13^3 = 4 \times_{15} 13 = 7$$

$$13^4 = 7 \times_{15} 13 = 1$$

$$13^5 = 1 \times_{15} 13 = 13$$

$\vdots$

For 14,

$$14^1 = 14$$

$$14^2 = 14 \times_{15} 14 = 1$$

$$14^3 = 1 \times_{15} 14 = 14$$

$\vdots$

Thus, there are no any generating elements in the given group. Therefore, the given group is not cyclic.